Spanish mackerel (*Scomberomorus maculatus*)

Summary Scores

- **Sensitivity:** Low
- **Exposure:** Moderate
- **Vulnerability:** Low

Sensitivity

Sensitivity Attributes	Score	LH DQ	Tally
Adult Mobility	1.16	2.6	
Complexity in Reproductive Strategy	1.12	2.1	
Dispersal of Early Life Stages	2	2.1	
Egg and Larval Survival and Settlement Requirements	1.52	2.06	
Historical Variability in Temperature	1.84	2.66	
Juvenile Habitat Requirements	1.4	2.2	
Other Stressors	2.4	2.4	
Population Growth Rate	1.44	2.7	
Prey Specificity	1.16	2.76	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	1.04	2.5	
Spawning Cycle	1.96	2.5	
Stock Size/Status	1.88	2.7	
Total Sensitivity | Certainty	**Low (1) | 1**		

Exposure

Exposure Factors	Mean	SD	Tally
Sea Surface Temperature	2.54	1.05	
Sea Surface Salinity	2.41	1.02	
Surface pH	3.71	0.74	
Mixed layer depth (delta rho = 0.03)	1.44	0.93	
Small zooplankton integrated biomass	1.08	0.25	
Medium zooplankton integrated biomass	1.43	0.60	
Large zooplankton integrated biomass	1.03	0.24	
Total Exposure	**Moderate (2.44)**	**0.91**	

Vulnerability

Vulnerability	Mean	SD	Tally
Total Vulnerability	**Low (2.44)**	**0.91**	

Low Moderate High Very High

Figure 1: Summary Sensitivity, Exposure, and Vulnerability Scores and Score Distributions for Spanish mackerel. Exposure and Vulnerability scores are averaged spatially across the species distribution and within stocks if applicable (see Figure 5 and Figure 4). Life History Data Quality (LH DQ) and Standard Deviations (SD) are reported for senstivity and exposure/vulnerability scores, respectively.

Changes from Previous Assessment(s)

Table 1: Changes to Vulnerability, Exposure, and Sensitivity scores relative to Previous Assessments. If a citation for the previous assessment is not provided, it comes from Hare et al. (2016).

Metric	Previous Assessment(s)	Current Assessment	Change
Sensitivity	Low	Low	None
Exposure	Very High	Moderate	Decrease
Vulnerability	Moderate	Low	Decrease
Directional Effect	Positive	Neutral	Decrease

Stocks

Spanish mackerel is managed as a single stock or stock information was not considered for this iteration of the CVA.

Score Descriptions:

Sensitivity:

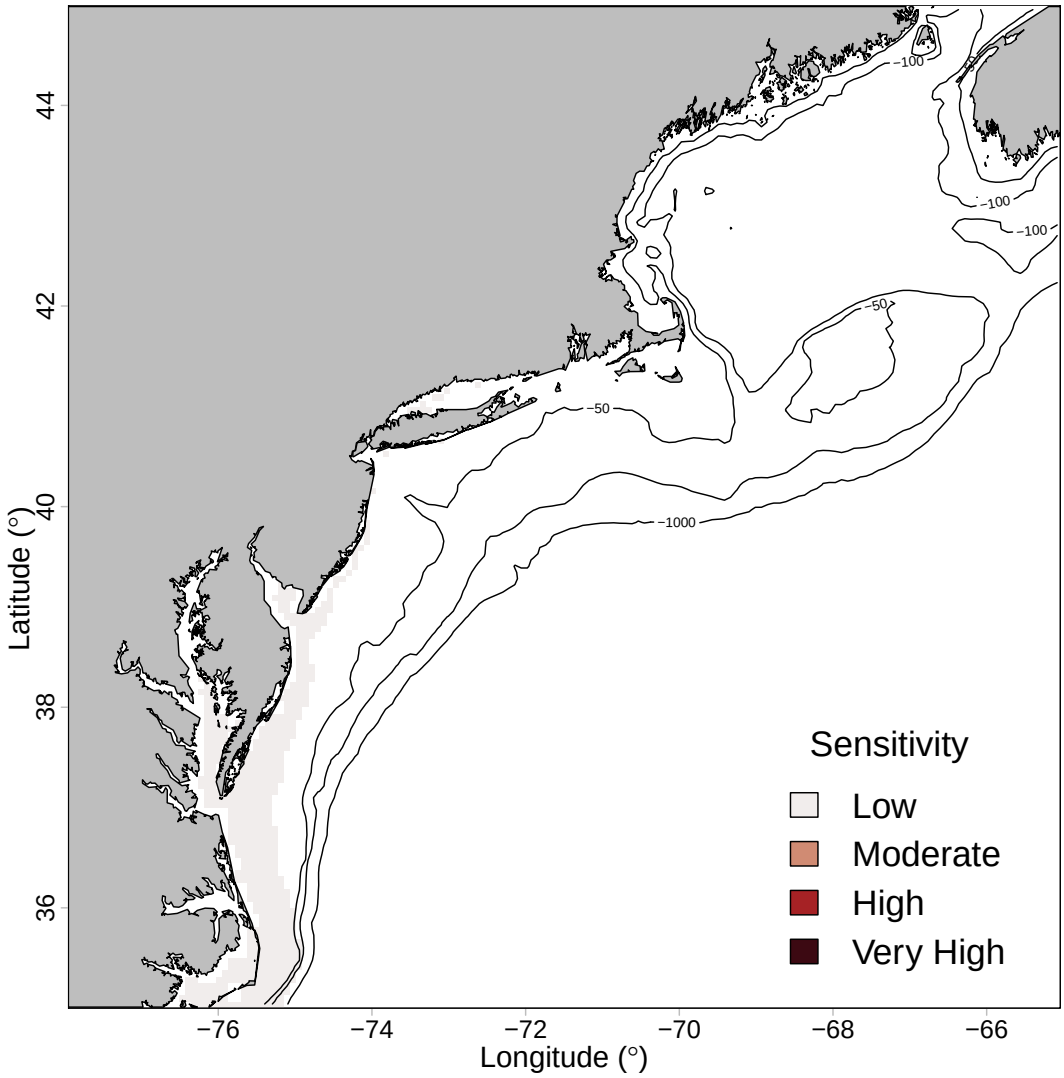


Figure 2: Spanish mackerel Sensitivity on the Northeast Shelf

Spanish mackerel received a **Low** sensitivity score. Certainty was 100% according to bootstrap analysis (Figure 1). No attributes scored higher than 2.5. The highest attribute score was for Other Stressors (2.4) (Figure 1). Spanish mackerel adults use estuarine water column habitats, which are classified as highly vulnerable due to impacts of stratification, hypoxia, increased fresh water runoff, and habitat loss (Guyant et al. 2022).

Similarly, in the previous Climate Vulnerability Assessment, Spanish Mackerel was also given a Low sensitivity score and no attributes scored above 2.5 (Hare et al. 2016).

Exposure:

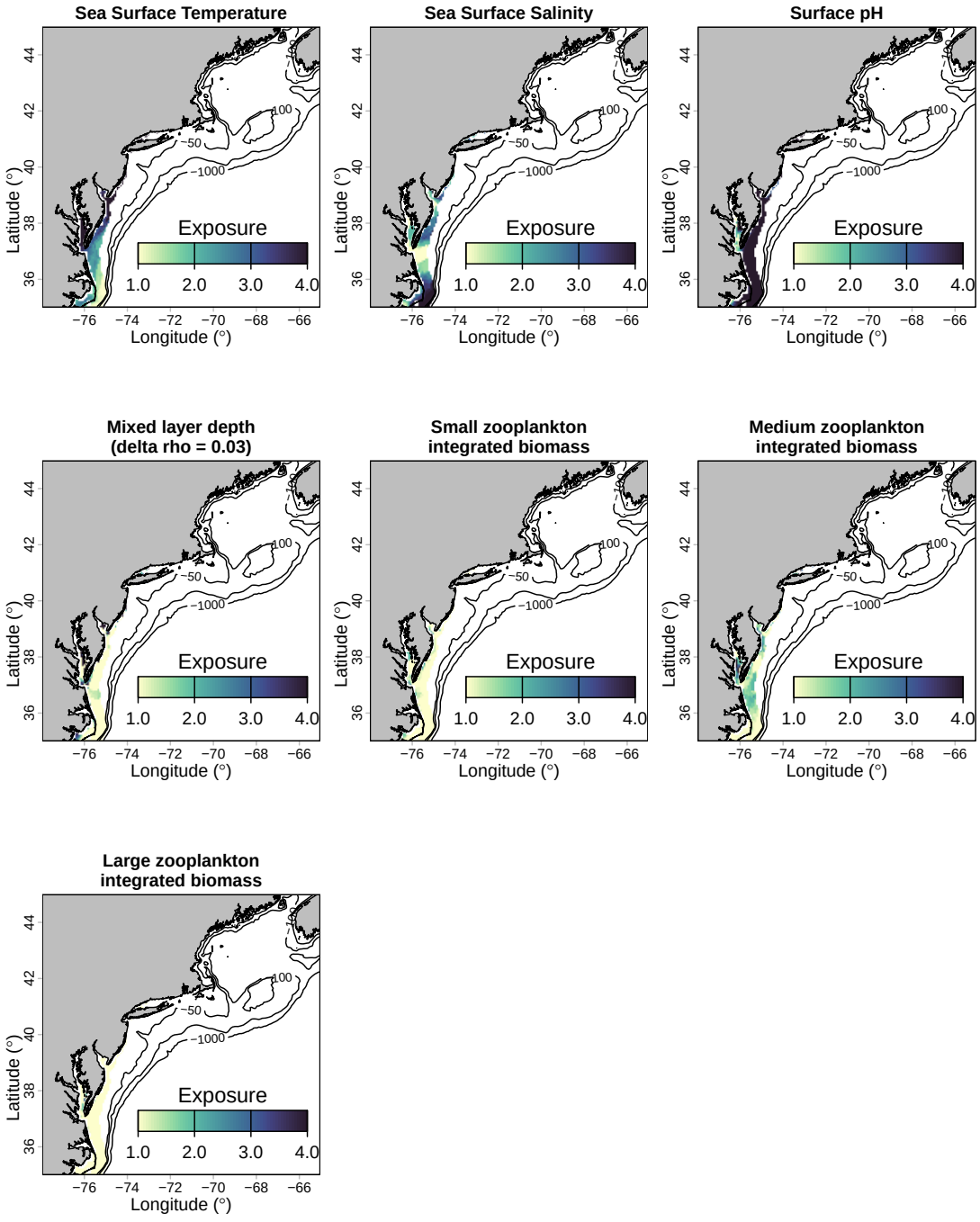


Figure 3: Maps of each Exposure Factor for Spanish mackerel. Exposure was calculated between 2014-2023 and 2025-2035

Spanish mackerel had an average Exposure score of **Moderate** (2.44 +/- 0.91) (Figure 1). Exposure was High in Chesapeake Bay and off the coasts from New Jersey to North Carolina, with the exception of the

mouth of Chesapeake Bay, where exposures were Low (Figure 4).

In the previous Assessment, Spanish mackerel received a Very High Exposure score, driven by high sea surface temperature, air temperature and ocean acidification exposure scores (Hare et al. 2016).

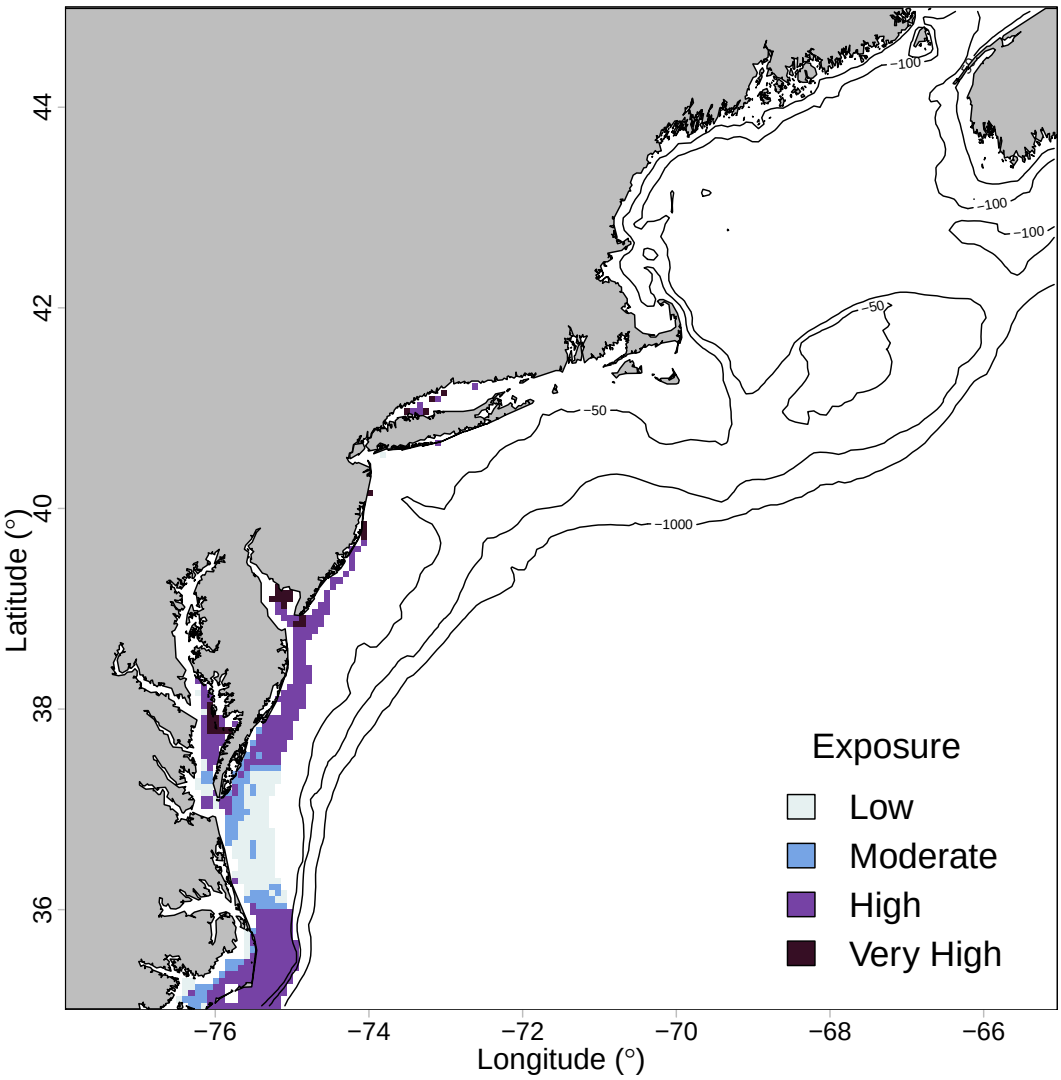


Figure 4: Total Exposure on the Northeast Shelf for Spanish mackerel. Total Exposure is calculated by applying the logic rule from Hare et al. (2016) within grid cells across exposure factors (see Figure 3)

Vulnerability:

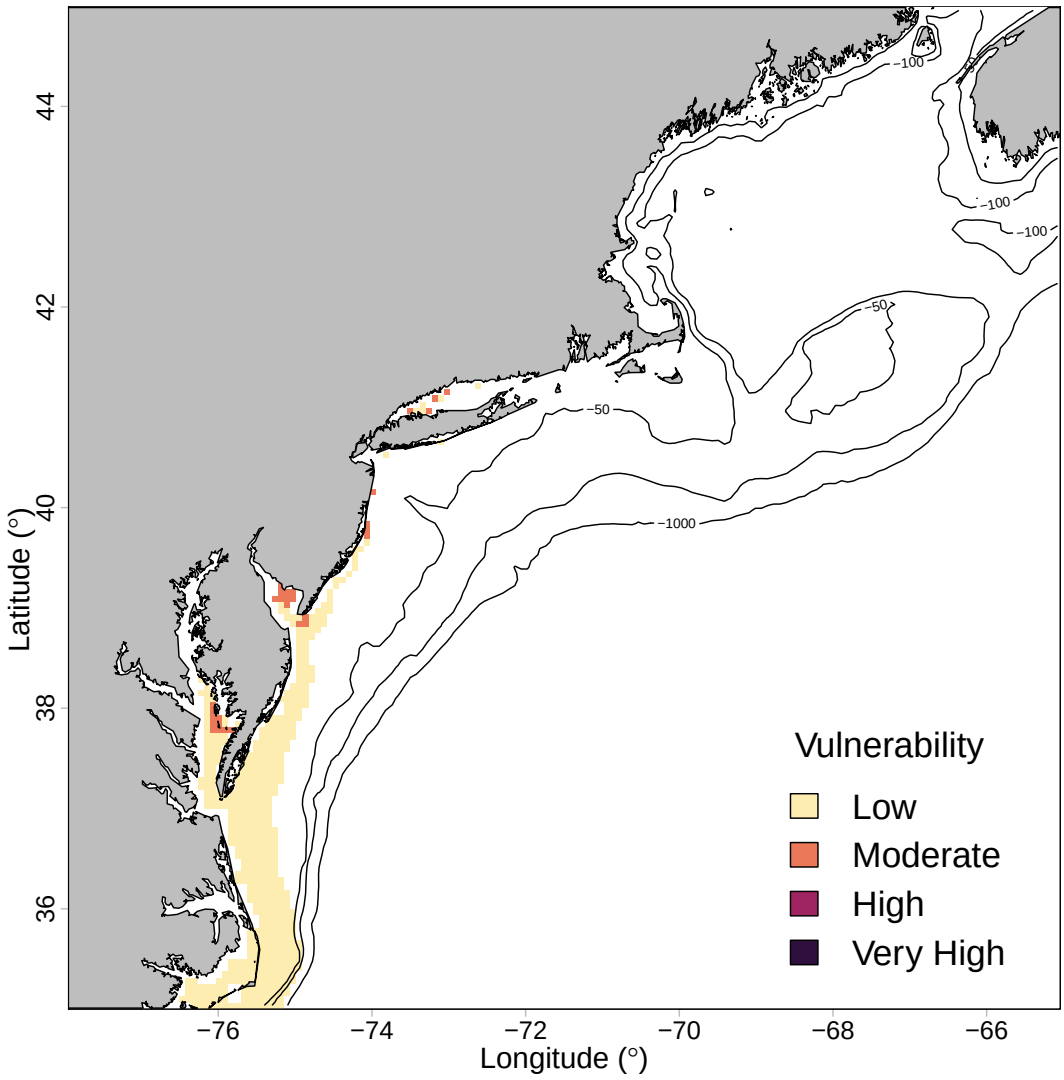


Figure 5: Vulnerability on the Northeast Shelf for Spanish mackerel. Vulnerability is the product of sensitivity and exposure (see Figure 4 and Figure 2)

Mean Vulnerability for Spanish Mackerel was **Low** (2.44 +/- 0.91) (Figure 1). There was little variation throughout their range (Figure 5).

In the previous Assessment, Spanish Mackerel received a Moderate score (Hare et al. 2016).

Data Quality Metrics

Life History Data Quality:

85% of data quality scores were greater than or equal to 2, indicating strong data quality. No data gaps (mean attribute data quality score < 2) were identified (Figure 1).

Model Confidence Score:

Mean model confidence score for spanish mackerel was 2.15 (+/- 0.22) indicating moderate confidence in the ensemble model. Experts stated that predicted distributions met expectations but low sample size decreased scores. The model AUC score was 0.99 indicating that the model had excellent predictive performance.

Additional Metrics

Distribution Shifts:

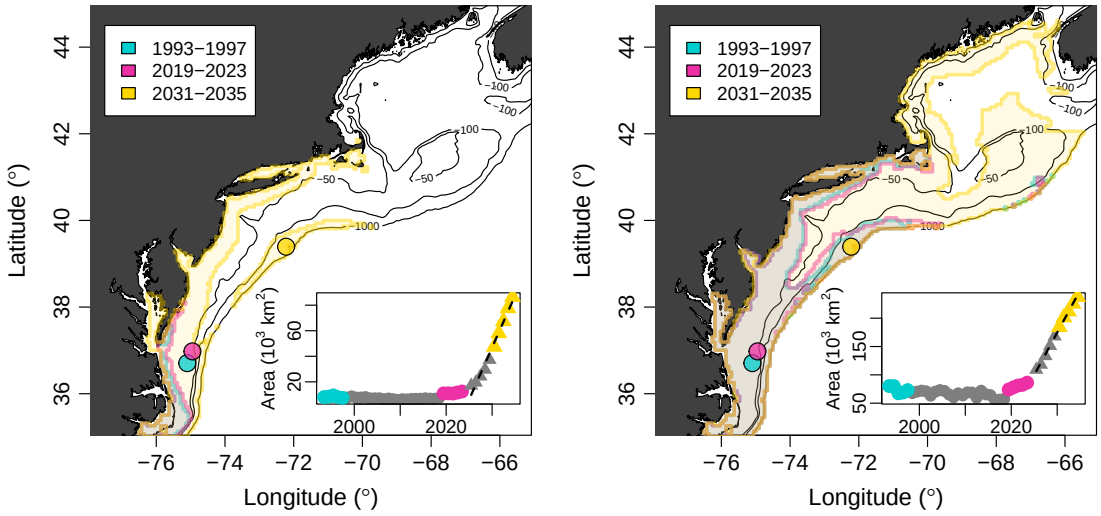


Figure 6: Distribution shifts on the Northeast Shelf for Spanish mackerel. Polygons illustrate core habitat (left) and species range (right) for five-year average periods. Points illustrate weighted centroid, calculated as the average position with species distribution models as weights. Inset timeseries illustrate the shift in core and habitat range area over time. Circles represent points from the MOM6 hindcast; triangles represent points from the MOM6 forecast. Dashed lines represent significant linear regressions

Note: For an animated version of this plot, please see the online version of this document.

Spanish mackerel remained stable between 1993 and 2023. The northern edge of the core habitat and overall species distribution moved northward, resulting in a shift in the weighted centroid 33 km to the north-northeast (Figure 6).

Spanish mackerel are expected to significantly expand their core habitat and range throughout the Northeast Shelf by 2035. Core habitat is predicted to expand from the coast of Maryland to south of Cape Cod. Similarly, Spanish mackerel are expected to expand their range to the coast of Gulf of Maine and Georges Bank (Figure 6). As a result, the weighted centroid is expected to shift 360 km northeast.

In the previous Assessment, spanish mackerel received a High Distributional Vulnerability rank due to their habitat preferences, seasonal migrations, and broad larval dispersal (Hare et al. 2016).

Table 2: Distribution shifts based on five-year average distributions. Changes are calculated relative to the previous time period.

Time Period	Weighted Centroid Shift (km & bearing)	Core Habitat Area (change) (km^2)	Habitat Total Area (change) (km^2)	Latitude Range (change) (deg)	Change in Longitude Range (deg)
1993-1997	NA	8,329	75,434	5.36	3.93
2019-2023	32.9 km NNE	11,404 (+3,075)	82,357 (+6,923)	5.61 (+0.25)	4.41 (+0.48)
2031-2035	359.5 km NE	69,287 (+57,883)	219,083 (+136,726)	7.86 (+2.24)	9.22 (+4.81)

Directional Effect:

Experts predict that climate change will have a **neutral** effect on Spanish mackerel. Certainty was 92%. Spanish mackerel prefer temperatures between 21-27 C, so warming temperatures will likely expand suitable habitat (Berrien and Finan 1977). Spawning, however, occurs in coastal areas and estuaries such as Chesapeake Bay (Cooksey 1996) (McEachran, Finucane, and Hall 1980) These habitats are vulnerable due to anthropogenic impacts and may impact reproductive success (Guyant et al. 2022). The previous Climate Vulnerability Assessment predicted that Spanish mackerel would be positively impacted by climate change (Hare et al. 2016).

Climate Effects on Abundance and Distribution

Little work exists on the effects of climate on Spanish mackerel populations in the Northwest Atlantic. Temperature likely limits movement; they are not found in waters colder than 18°C (Berrien and Finan 1977). Rising ocean temperatures may expand suitable habitat in the Mid-Atlantic and north to New England. Larval Spanish mackerel were largely resistant to contaminants from the Deepwater Horizon oil spill, and larval abundance did not differ before v. after the spill (Ransom, Filbrun, and Hernandez Jr 2016).

Life History Synopsis

Spanish mackerel are a migratory, schooling fish found along the eastern United States and Gulf from New York to Florida (Godcharles and Murphy 1986). There are two Spanish mackerel stocks, one in the South Atlantic and one in the Gulf of America (SouthEast Data, Assessment, and Review (SEDAR) 2022) (SouthEast Data, Assessment, and Review (SEDAR) 2023). The Atlantic stock overwinters off the Florida coast and migrates northward to the Mid-Atlantic in the spring and early summer. They return south to Florida once the Mid-Atlantic cools in the fall (Berrien and Finan 1977). Spanish mackerel prefer temperatures 21-27°C and are not seen below 18°C (Berrien and Finan 1977). Spawning occurs from May to September along the southeast Atlantic coast, with lower latitudes spawning earlier than higher latitudes (Collins and Stender 1987). In Chesapeake Bay, spawning occurs from June to August, peaking in June (Cooksey 1996). Females are batch spawners, releasing as many as 1.5 million eggs over the spawning season (Collins and Stender 1987). Larval collection data indicate that spawning occurs at depths of 12-35 meters over the inner continental shelf including estuarine and coastal areas (McEachran, Finucane, and Hall 1980). Hatching occurs approximately 25 hours after fertilization at water temperatures averaging 26ºC. Larvae and early juveniles grow 1.9 mm per day for approximately

the first 23 days of life. From 23-40 days, growth is accelerated, with young fishes growing as much as 5 mm per day. Thereafter, growth slows to approximately 2.1 mm per day (Schmidt, Collins, and Wyanski 1993). Larvae are found in surface waters between 19.6–29.8°C with a salinity of 28.3–37.4 ppt. It is unclear if there is a dependency on currents for transport to nursery habitats (Godcharles and Murphy 1986). Larvae feed on other larval fishes such as carngids, clupeids, and engraulids and crustaceans (Berrien and Finan 1977). Juveniles and adults are voracious, opportunistic, carnivores. Food consists mostly of small fishes with lesser quantities of pandalid and penaeoid shrimps and squids. Clupeoids such as round herring, thread herring, menhaden, and alewives, along with anchovies, are important forage species. They seldom feed on bottom invertebrates (Bowman et al. 2000) (Berrien and Finan 1977). Spanish mackerel reach maturity around 1-1.5 years of age (Juan-Jord'a et al. 2013) with a maximum lifespan of around 12 years (SouthEast Data, Assessment, and Review (SEDAR) 2022). They are largely preyed upon by dolphins, sharks, and tunas (Berrien and Finan 1977). Management of Spanish mackerel in state waters is coordinated by the Atlantic States Marine Fisheries Commission (ASMFC) Coastal Pelagics Management Board. Both stocks are not overfished or subject to overfishing (SouthEast Data, Assessment, and Review (SEDAR) 2022) (SouthEast Data, Assessment, and Review (SEDAR) 2023).

Supplemental Scores

Exposure Timeseries

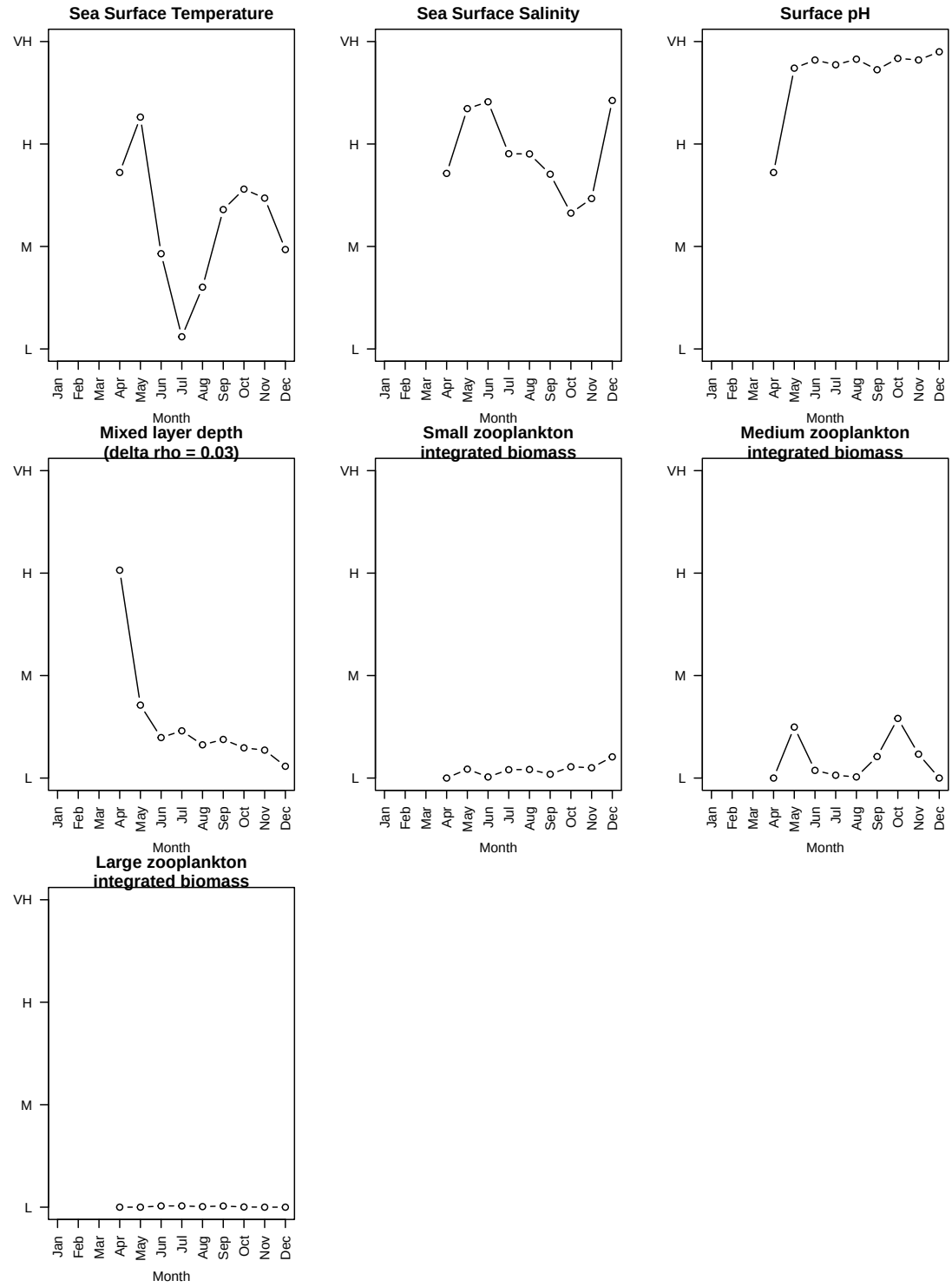


Figure 7: Monthly Timeseries of each Exposure Factor for Spanish mackerel. Exposure was calculated between 2014-2023 and 2025-2035

Exposure scores ranged from Low to High across months, with High scores in May, June, and December (Figure 8). These patterns were driven primarily by sea surface salinity and surface pH exposures (Figure 7).

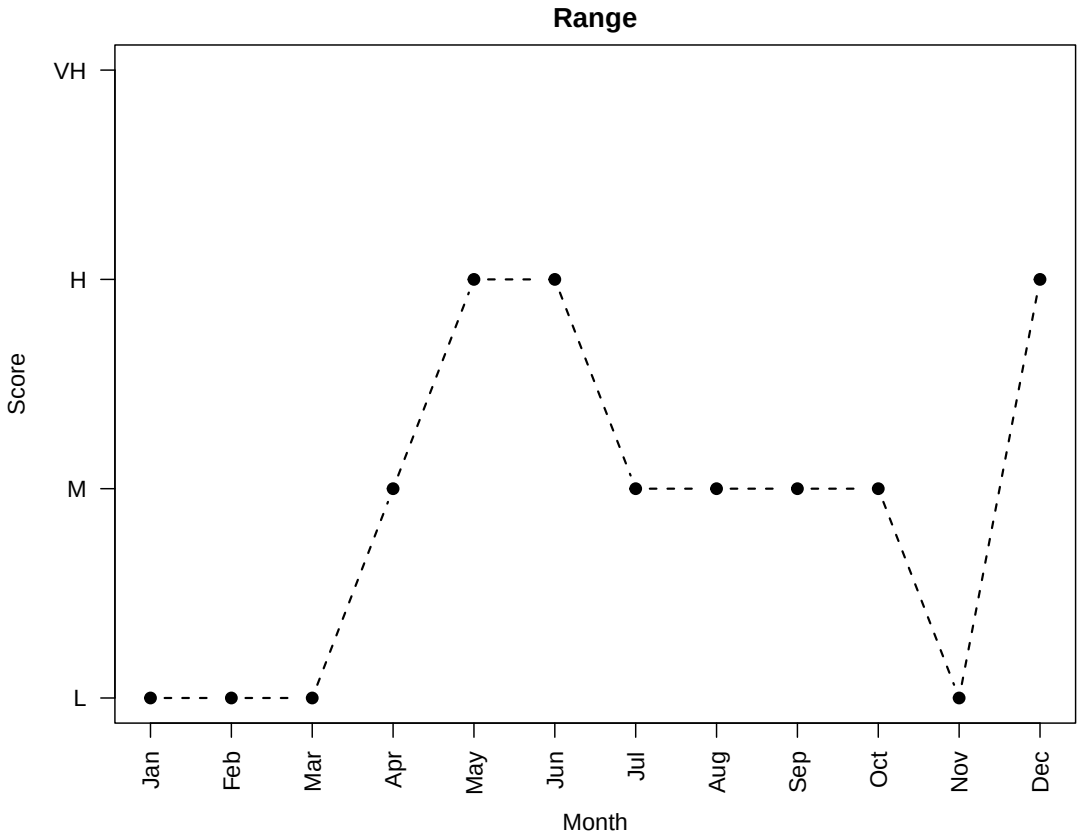


Figure 8: Timeseries of Total Exposure for Spanish mackerel

Exposure due to Significant Factors

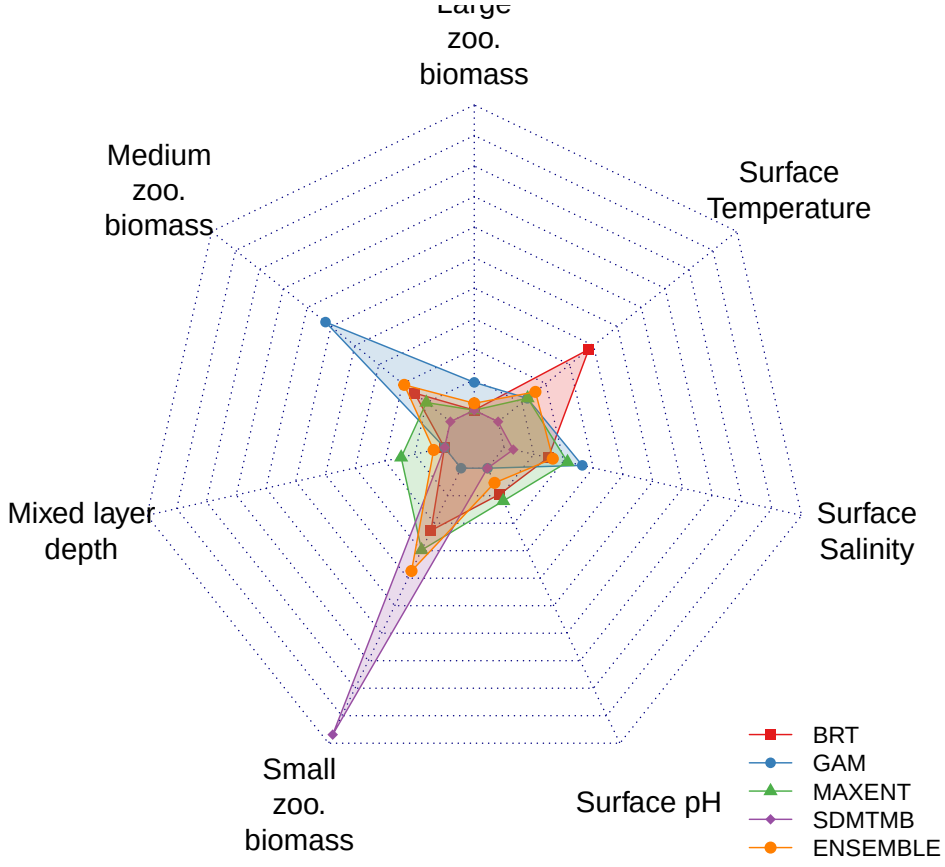


Figure 9: Dynamic Factors driving Spanish mackerel distributions. The closer the polygon is to the edge of the figure, the more important the variable is. Each ring represents 10% variance explained. Factors must explain at least 10% of variance in the ensemble model to be used to calculate total exposure.

Sea surface temperature and salinity, as well as small and medium zooplankton biomass were important dynamic drivers of species distributions (Figure 9). Spanish mackerel prefer warmer temperatures (Berrien and Finan 1977). They are generalist predators on smaller fishes (Bowman et al. 2000) (Berrien and Finan 1977). When these factors were used to calculate exposure, the mean exposure score decreased to Low (1.3 +/- 0.62) (Figure 11). These scores were greatest in Chesapeake Bay and off the coast of Maryland and Virginia (Figure 10).

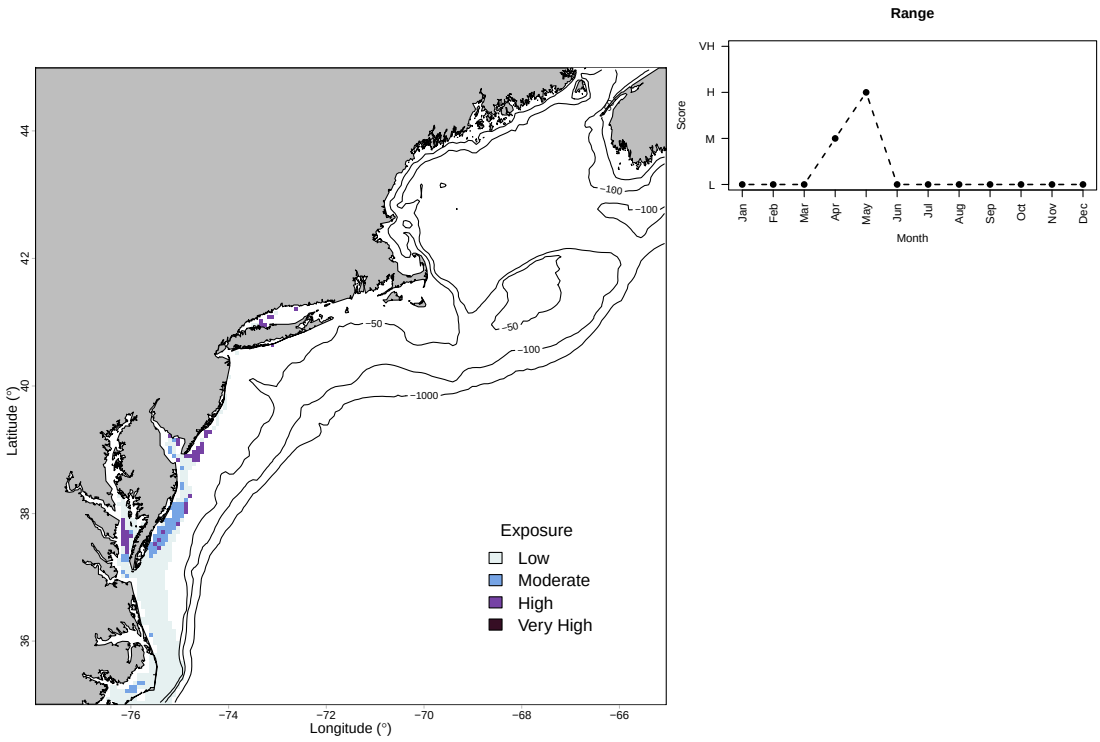


Figure 10: Total Exposure Map and Timeseries when Exposure was calculated with only Exposure Factors critical to Spanish mackerel distributions

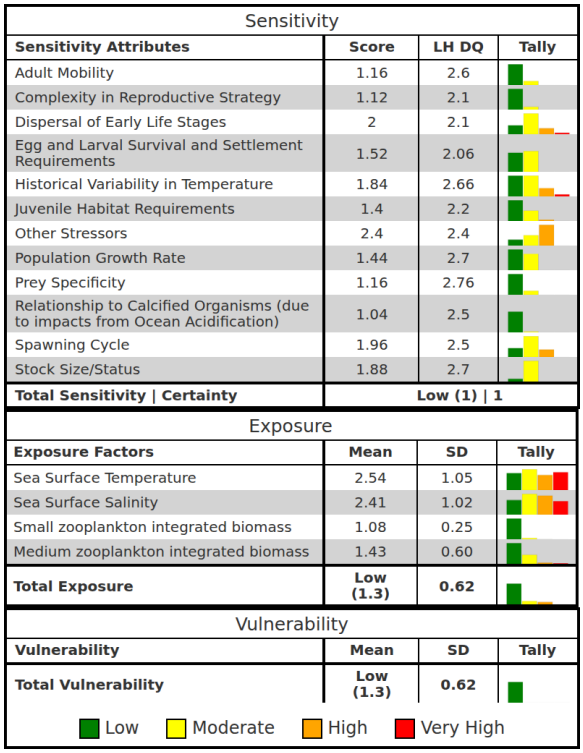


Figure 11: As in Figure 1 for Spanish mackerel, but with only significant Exposure factors and the resulting Vulnerability Scores.

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